

BCPS 422: Robotics and Embedded Control Systems

Chapter 1: Introduction to Embedded Systems

Introduction

- Computers are no longer limited to desktop and laptop machines.
- Today, computers are embedded in everyday objects.
- Examples:

-  **Washing Machines**
-  **Microwave Ovens**
-  **Cars**
-  **Smart Watches**
-  **Drones**
-  **Medical Devices**
-  **Industrial Robots**

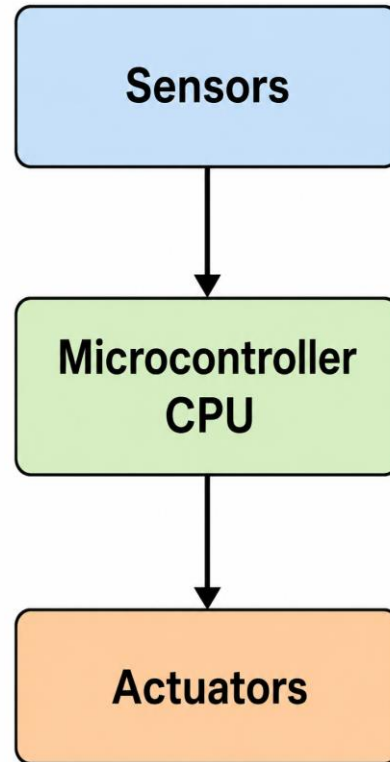
Introduction (cont)

- These specialized computers are called embedded systems because they are embedded into larger devices to perform dedicated functions

Definition of Embedded Systems

- An embedded system is a combination of hardware and software designed to perform a dedicated function within a larger system
- Unlike general-purpose computers, embedded systems are optimized for specific tasks
- A microprocessor-based system designed to control, monitor, or assist the operation of equipment, machinery, or devices.

Block Diagram of an Embedded System



- Sensors gather data, CPU processes data, Actuators perform actions.

Description: Sensors

- Gather information from the environment.
- Examples:
 - Temperature sensors
 - Light sensors
 - Pressure sensors
 - Ultrasonic sensors

Description: Processor

- The processor interprets sensor data and makes decisions.
- Examples:
 - ARM Cortex processors
 - AVR processors
 - ESP32
 - Raspberry Pi

Description: Actuators

- Act upon processor commands.
- Examples:
 -  Motors
 -  Solenoids
 -  Valves
 -  Relays

Real-World Example

Automatic Water Tank Controller

- Inputs
 - Water level sensor
- Processor
 - Arduino Uno
- Outputs
 - Water pump
- Operation:
 - Sensor measures water level.
 - Arduino reads sensor value.
 - If level falls below threshold:
 - Pump turns ON.
 - If tank becomes full:
 - Pump turns OFF.

Characteristics of Embedded Systems

- Embedded systems possess several distinguishing features.

1. Dedicated Functionality

- They are designed for specific tasks.
- Examples:
 - Real-Time Operation

Characteristics of Embedded Systems (cont)

2. Responses must occur within strict timing constraints

—Examples:

- Airbag systems
- Flight controllers
- Industrial robots

Characteristics of Embedded Systems (cont)

3. Many embedded systems operate continuously for years

—Examples:

- Traffic lights
- Telephone exchanges
- Power stations

Characteristics of Embedded Systems (cont)

4. Low Power Consumption

- Battery-operated systems require efficient energy usage
- Examples:
 - Smart watches
 - IoT devices
 - Wireless sensors

Characteristics of Embedded Systems (cont)

5. Compact Size

- Embedded systems are often miniaturized.
- Examples:
 - Medical implants
 - Smartphones
 - Hearing aids

Classification of Embedded Systems

- Standalone Embedded Systems
- Real-Time Embedded Systems
- Networked Embedded Systems
- Mobile Embedded Systems

Standalone Embedded Systems

- Operate independently
- Examples:
 - Calculator
 - Digital camera
 - Microwave oven

Real-Time Embedded Systems

- Must satisfy timing requirements.
 - **Hard Real-Time Systems**
 - **Soft Real-Time Systems**

Hard Real-Time Systems

- Missing deadlines causes catastrophic failure.
- Examples:
 - Aircraft autopilot
 - Nuclear plant controller

Soft Real-Time Systems

- Occasional delays are acceptable
- Examples:
 - Video streaming
 - Multimedia systems

Networked Embedded Systems

- Connected through communication networks.
 - Examples:
 - Smart homes
 - IoT devices
 - Remote monitoring systems

Mobile Embedded Systems

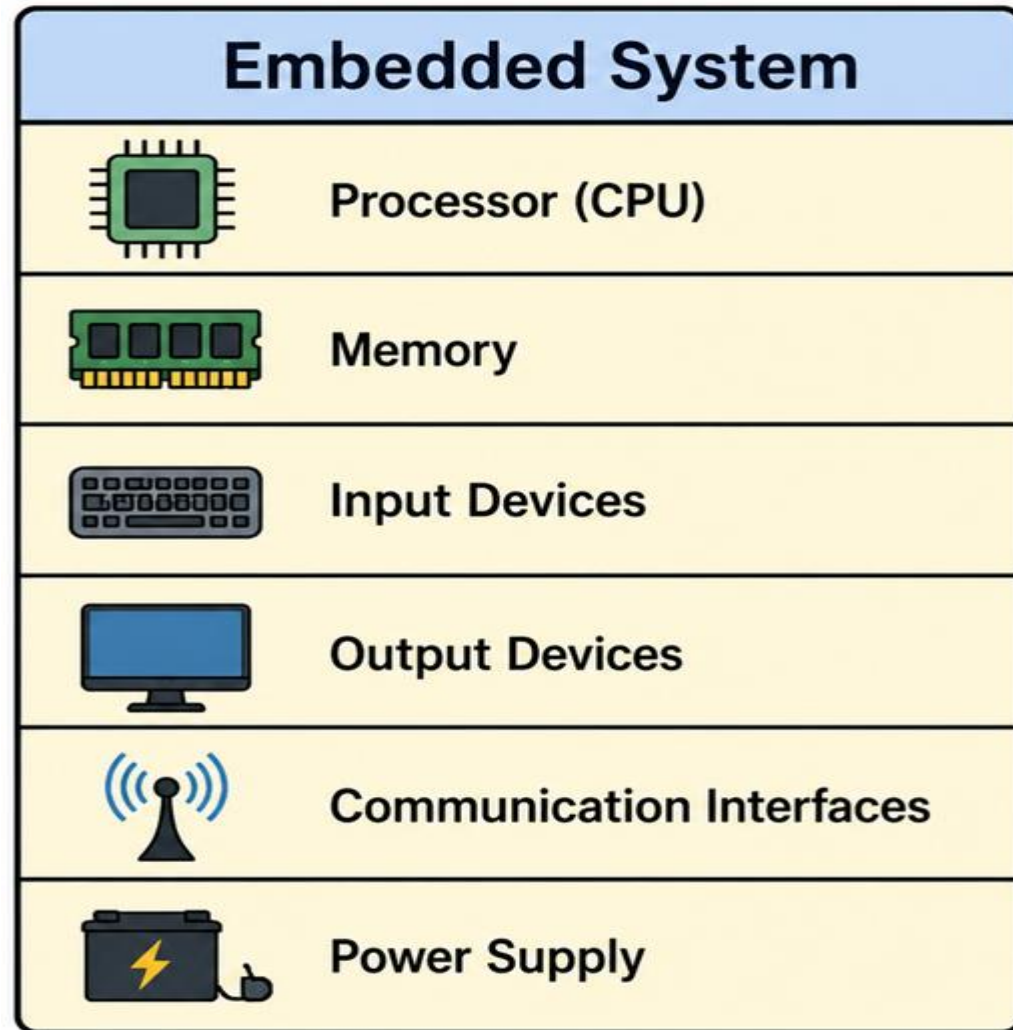
- Portable embedded devices.
- Examples:
 - Smartphones
 - Tablets
 - GPS devices

Embedded Systems vrs General Computer

Feature	Embedded System	General Computer
Purpose	Specific task	Multiple tasks
Hardware	Optimized	General
Power Consumption	Low	High
Cost	Lower	Higher
Flexibility	Limited	High
Operating System	RTOS	Windows/Linux

Architecture of Embedded Systems

- An embedded systems contain



Processor

- The processor executes instructions.
- Examples:
 - ATmega328P
 - ARM Cortex-M4
 - ESP32

Memory

- **RAM**
 - Stores temporary data.
- Characteristics:
 - Volatile
 - Fast
- **ROM/Flash**
 - Stores programs permanently.
- Characteristics:
 - Non-volatile
 - Retains data without power

Input Devices

- Examples:
 - Switches
 - Keypads
 - Sensors

Output Devices

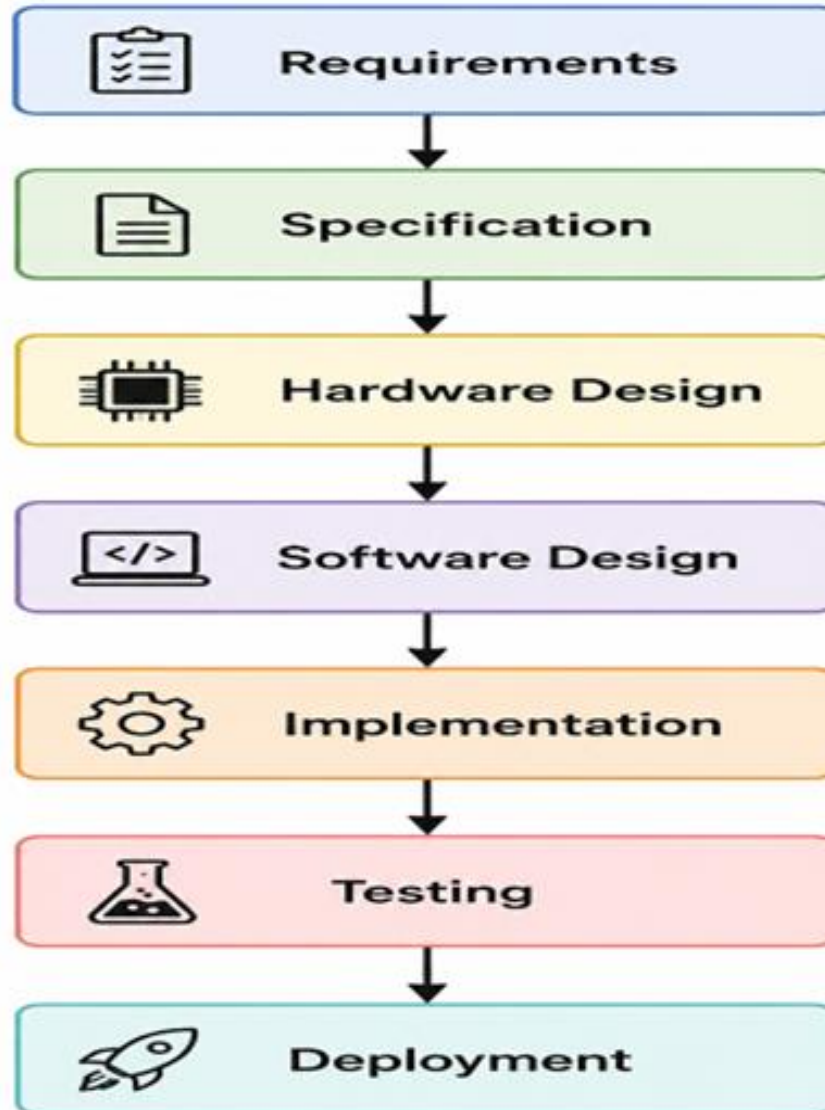
- Examples:
 - LEDs
 - Motors
 - LCD displays

Communication Interfaces

- Examples:

Interface	Use
UART	Serial communication
SPI	High-speed peripherals
I²C	Multiple devices
CAN	Automotive systems
Ethernet	Networking

Embedded System Design Process



Steps in Developing an Embedded system

- **Step 1: Requirement Analysis**
 - Determine:
 - What should the system do?
 - Environmental conditions
 - Cost constraints

Steps in Developing an Embedded system (cont)

- **Step 2: System Specification**
 - Specify:
 - Inputs
 - Outputs
 - Performance

Steps in Developing an Embedded system (cont)

- **Step 3: Hardware Design**
 - Select:
 - Processor
 - Sensors
 - Actuators

Steps in Developing an Embedded system (cont)

- **Step 4: Software Development**
 - Choose programming language:
 - C
 - C++
 - Python
 - MATLAB

Steps in Developing an Embedded system (cont)

- **Step 5: Testing**
 - Testing includes:
 - Unit testing

Applications of Embedded Systems

- **Consumer Electronics**

- Examples:

- Smart TVs

- Cameras

- Smartphones

Applications of Embedded Systems (cont)

- **Healthcare**
 - Examples:
 - MRI machines
 - ECG monitors
 - Pacemakers

Applications of Embedded Systems (cont)

- **Transportation**
 - Examples:
 - ABS braking systems
 - Cruise control
 - Autonomous vehicles

Applications of Embedded Systems (cont)

- **Industrial Automation**
 - Examples:
 - PLC controllers
 - CNC machines
 - Industrial robots

Applications of Embedded Systems (cont)

- Aerospace
 - Examples:
 - Flight control systems
 - Satellite controllers
 - UAVs

ASSIGNMENT

Question 1

- A hospital intends to deploy a patient monitoring system that continuously measures heart rate and oxygen levels. Analyze the characteristics of embedded systems that make them suitable for this application.

ASSIGNMENT (CONT)

Question 2

- Analyze the consequences of removing sensors from an embedded system.

Question 3

- An autonomous vehicle misses a braking deadline by 0.5 seconds. Evaluate the type of embedded system involved and the implications.